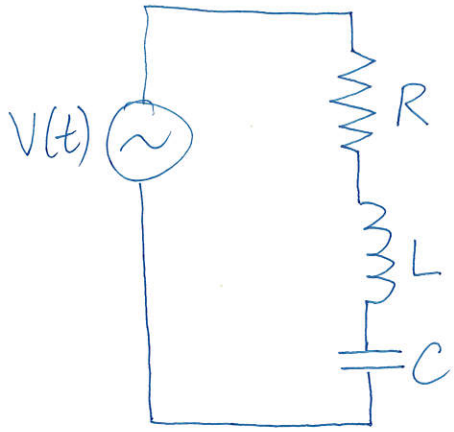
AC circuits: resistors, inductors, capacitors



Voltage drop

IR

$-\left(-L\frac{dI}{dt}\right) = L\frac{dI}{dt}$

$\frac{q}{C} = \frac{1}{C}\int_0^t I\,dt$ if $I(t=0)=0$.

$$\Rightarrow \quad V = IR + L\frac{dI}{dt} + \frac{1}{C}\int_0^t I\,dt$$

Look for ac-solution: $V(t) = V_0 e^{i\omega t}$, $I(t) = I_0 e^{i\omega t}$ (phys. quantity is Real part of that)

$$\Rightarrow \quad V_0 e^{i\omega t} = I_0 e^{i\omega t} R + L i\omega I_0 e^{i\omega t} + \frac{I_0 e^{i\omega t}}{i\omega C}$$

$$\Rightarrow \begin{cases} V_0 = I_0\left(R + i\omega L + \frac{1}{i\omega C}\right) = I_0 Z \\ V(t) = Z\, I(t) \end{cases}$$

Z ↳ impedance (complex quantity!)

Power absorbed by the circuit:

$P(t) = \langle V(t)\, I(t) \rangle$ ↳ averaged over the period of oscillation

$$I(t) = \frac{V(t)}{Z}$$

$$Re\left(I(t)\right) = Re\left(\frac{V(t)}{Z}\right) = Re\left(\frac{V' + iV''}{Z' + iZ''}\right) =$$

$$= Re\left(\frac{(V' + iV'')(Z' - iZ'')}{(Z')^2 + (Z'')^2}\right) = \left(\frac{V'Z' + V''Z''}{|Z|^2}\right)$$

$$V' = V_0 \cos\omega t \ , \ V'' = V_0 \sin\omega t$$

$$Z = R + i\omega L - \frac{i}{\omega C} \Rightarrow Z' = R \ , \ Z'' = \omega L - \frac{1}{\omega C}$$

$$Re\left(I(t)\right) = \frac{V_0 \cos\omega t \, R + V_0 \sin\omega t \left(\omega L - \frac{1}{\omega C}\right)}{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

<u>Simpler</u> let $Z = |Z| e^{i\varphi} \Rightarrow Z' = |Z| \cos\varphi \ , \ Z'' = |Z| \sin\varphi$

then $Re\left(I(t)\right) = \dfrac{V_0 |Z| \left(\cos\omega t \cos\varphi + \sin\omega t \sin\varphi\right)}{|Z|^2} = \dfrac{V_0 \cos(\omega t - \varphi)}{|Z|}$

$$= |I_0| \cos(\omega t - \varphi)$$

↳ phase delay

$$|I_0| = \frac{V_0}{|Z|} = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

resonance at $\omega = \omega_0 = \frac{1}{\sqrt{LC}}$

Thus power $P = \frac{1}{T}\int_0^{T=2\pi/\omega} dt\ V_0 \cos\omega t\ I_0 \cos(\omega t - \varphi)$

$$= \frac{\omega}{2\pi}\int_0^{2\pi} \frac{d(t\omega)}{\omega}\ V_0 I_0\ \frac{1}{2}\left(e^{i\omega t} + e^{-i\omega t}\right)\frac{1}{2}\left(e^{i\omega t - i\varphi} + e^{-i\omega t + i\varphi}\right)$$

$$= \frac{1}{2\pi}\int_0^{2\pi} dx\ V_0 I_0\ \frac{\left(e^{ix} + e^{-ix}\right)}{2}\ \frac{e^{ix}e^{-i\varphi} + e^{-ix}e^{i\varphi}}{2}$$

$$= \frac{1}{8\pi} V_0 I_0 \int_0^{2\pi} dx \left(e^{i2x}e^{-i\varphi} + e^{i\varphi} + e^{-i\varphi} + e^{-i2x}e^{i\varphi}\right)$$

$$\int_0^{2\pi} dx\ e^{i2x} = \frac{1}{2i} e^{i2x}\Big|_0^{2\pi} = 0$$

$$= \frac{1}{8\pi} V_0 I_0\ 2\cos\varphi \cdot 2\pi = \frac{1}{2} V_0 I_0 \cos\varphi.$$

Since $V_0 = |I_0||Z|$, $P = \frac{1}{2}\underbrace{|Z|\cos\varphi}_{Z'}\ I_0^2 =$

$$= \frac{1}{2}\,\mathrm{Re}(Z)\, I_0^2.$$

Energy is dissipated only by $\mathrm{Re}(Z) = R$.

Transmission lines

Key assumption for AC circuit: $V(t)$ (and $I(t)$) has the <u>same phase</u> throughout the circuit.

This is true if the wavelength of the oscillation

$$\lambda = \frac{2\pi c}{\omega} \gg \text{size of the circuit.}$$

Land (telephone) line: $\nu = 56\ \text{kHz}$

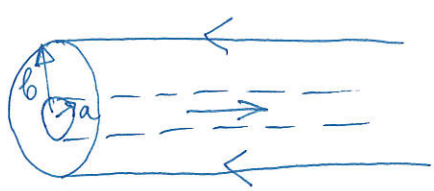
$$\lambda = \frac{c}{56 \cdot 10^3\ s^{-1}} = \frac{3 \cdot 10^8\ m/s}{56 \cdot 10^3\ 1/s}$$

$$\cong \frac{30}{60} \cdot 10^4\ m = 5 \cdot 10^3\ m = 5\ km$$

Computer network: $\nu \sim 100\ \text{MHz} \simeq 100 \cdot 10^6\ \text{Hz}$

$$\lambda = \frac{3 \cdot 10^8\ m/s}{10^8\ 1/s} = 3\ m\ (!)$$

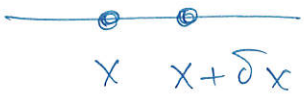
$\Rightarrow$ the phase of $V(t, x)$ is different at different positions $\underline{x}$ at the same instant $\underline{t}$.



Parallel conductors

capacitance per unit length C

inductance per unit length L



1) $V(x,t) - V(x+\delta x,t) = L\,\delta x \dfrac{\partial I(x,t)}{\partial t}$

$$\Rightarrow \frac{\partial V(x,t)}{\partial x} = -L \frac{\partial I(x,t)}{\partial t}$$

2) $$\underbrace{\int_0^t d\tau \left(I(x,t) - I(x+\delta x,t)\right)}_{Q} = C\,\delta x\, \underbrace{\delta V(x,t)}_{\text{pot. difference between } x \text{ and } \delta x}$$

$$-\int_0^{\Delta t} dt \left(\frac{\partial I(x,t)}{\partial x}\right)\delta x = C\,\delta x\,\delta V(x,t)$$

$t \to 0$

$$\Rightarrow \frac{\partial I(x,t)}{\partial x} = -C \frac{\partial V(x,t)}{\partial t}$$

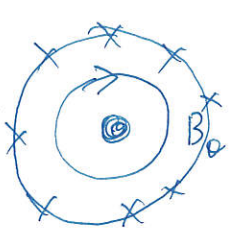
$$\left.\begin{aligned} \frac{\partial^2 V}{\partial x^2} &= -L \frac{\partial^2 I}{\partial x \partial t} \\ \frac{\partial^2 I}{\partial t \partial x} &= -C \frac{\partial^2 V}{\partial t^2} \end{aligned}\right\} \quad \boxed{\frac{\partial^2 V}{\partial x^2} = LC \frac{\partial^2 V}{\partial t^2}}$$

wave equation

$$v = \frac{1}{\sqrt{LC}}$$

Coaxial cable: $C = \dfrac{2\pi\varepsilon_0}{\ln(b/a)}$

Inductance



$$B_\theta = \frac{\mu_0 I}{2\pi r} \Rightarrow \Phi = \int_a^b B_\theta\, dr = \frac{\mu_0 I}{2\pi} \ln(b/a)$$

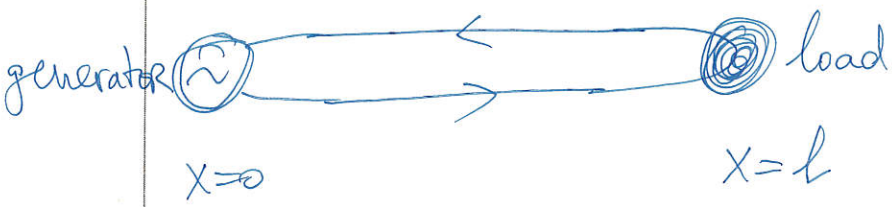
per unit length

$$\Rightarrow L = \frac{\Phi}{I} = \frac{\mu_0}{2\pi} \ln(b/a)$$

$$\Rightarrow v = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{2\pi\varepsilon_0}{\ln(b/a)} \frac{\mu_0 \ln(b/a)}{2\pi}}} = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = c$$

speed of light!

$$\frac{1}{v^2}\frac{\partial^2 V}{\partial t^2} = \frac{\partial^2 V}{\partial x^2}$$ wave eqn.



$V(0,t) = V_0 \cos(\omega t) \Rightarrow$ this is boundary condition

$$\Rightarrow \underbrace{V(x,t) = V_0 \cos(\omega t - kx)}_{\text{satisfies wave equation}} \quad \left(k = \frac{\omega}{v}\right)$$

wave propagating from generator to the load.

$$\frac{\partial^2 V(x,t)}{\partial x^2} = -k^2 V_0 \cos(\omega t - kx)$$

$$\frac{\partial^2 V(x,t)}{\partial t^2} = -\omega^2 V_0 \cos(\omega t - kx)$$

$$\Rightarrow \frac{1}{v^2}\frac{\partial^2 V}{\partial t^2} = -\underbrace{\frac{\omega^2}{v^2}}_{k^2} V_0 \cos(\omega t - kx) = \frac{\partial^2 V(x,t)}{\partial x^2}$$

!

Since $-L\frac{\partial I}{\partial t} = \frac{\partial V}{\partial x}$

$$L\frac{\partial I}{\partial t} = -\frac{\partial}{\partial x}\left(V_0 \cos(\omega t - kx)\right) =$$

$$= -k V_0 \sin(\omega t - kx)$$

$$\Rightarrow I(t) = A\cos(\omega t - kx)$$

$$\frac{\partial I}{\partial t} = -A\omega \sin(\omega t - kx)$$

$$\Rightarrow L\omega A = k V_0$$

$$A = \frac{k V_0}{L\omega} = \frac{V_0\, \omega/v}{L\omega} =$$

$$= \frac{V_0}{Lv} = \left(v = \frac{1}{\sqrt{LC}}\right)$$

$$A = \frac{V_0}{\sqrt{L/C}}$$

This solution makes sense if

$$Z(\ell) = \frac{V(\ell,t)}{I(\ell,t)} = \frac{V_0 \cos(\omega t - \ell x)}{\frac{V_0}{\sqrt{L/C}} \cos(\omega t - \ell x)} = \sqrt{\frac{L}{C}}$$

(This is load matching condition)

Input impedance $Z_{in} = \frac{V(0,t)}{I(0,t)} = \sqrt{\frac{L}{C}}$

What $Z(\ell) \neq Z_0 = \sqrt{\frac{L}{C}}$

$\Rightarrow$ some of the power is <u>reflected</u> back

$$\begin{cases} V(x,t) = \underbrace{V_0 e^{i(\omega t - kx)}}_{\text{wave moving right}} + \underbrace{\varkappa V_0 e^{i(\omega t + kx)}}_{\text{reflected wave: moves left}} \\ I(x,t) = \frac{V_0}{Z_0} e^{i(\omega t - kx)} - \frac{\varkappa V_0}{Z_0} e^{i(\omega t + kx)} \end{cases}$$

$\varkappa$ = reflection coefficient

$\varkappa \sim (R - Z_0)$ so that no reflection when load is matched.